Shahjalal University of Science and Technology, Sylhet

Department of Bangla

Research Paper

Course No: STA 225b

Course Title : Basic Statistic in Research

Research about *Socio-economic condition of Pharmaceutical Representative.*

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Objectives:

1. What is the Mean of total monthly income?
2. What is the Median of total monthly income?
3. What is the Mode of Monthly Phone Exp?
4. What is the Mode of total monthly income?
5. What is the 3rd Quartile Monthly Travelling Exp?
6. What is the 1st Quartile of Monthly Travelling Exp?
7. What is the Correlation between total income and total exp?
8. What is the Correlation between travelling exp. And Phone exp.? Comment.
9. What is the Correlation between Age and Phone Exp.? Comment.
10. What is the Mean and Standard deviation of monthly family expenditure of Pharmaceutical Representative?

Table 01: Categorical Variable

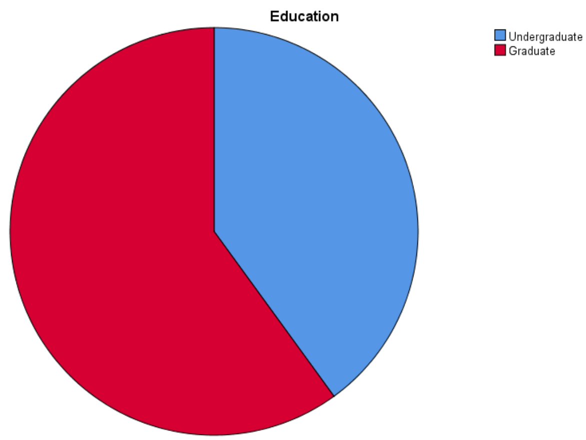
Characteristics	Number of Representative	Percentage
Education		
• Undergraduate	04	40
• Graduate	06	60
Marital Status		
• Unmarried	07	70
• Married	03	30

The following table deals with the Categorical Variable of 10 Pharmaceutical Representative's educational and marital status. There are 40% of the Pharmaceutical Representative is Undergraduate and 60% are Graduate. The marital status of the Pharmaceutical Representatives, among them 70% of the Pharmaceutical Representatives are unmarried and 30% are married.

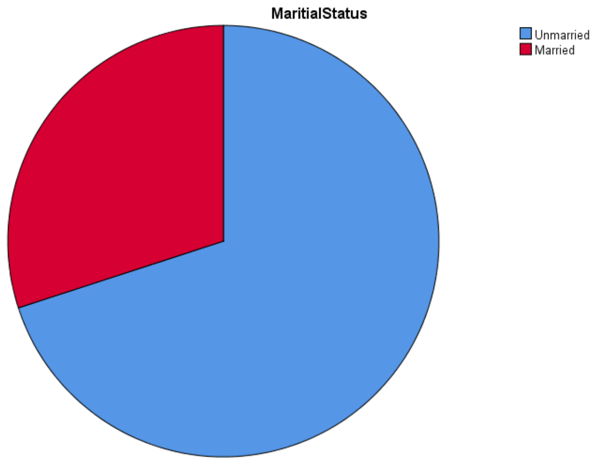
Graph: Pie-Chart, Bar Chart

Pie-Chart

The Pie-Chart consists of a circle sub-divided by into sectors, whose areas are proportional to the various parts into which the whole quantity is divided.



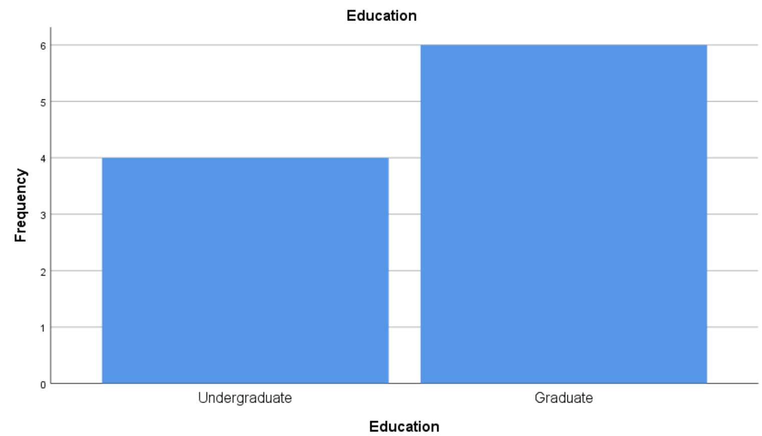
The Pie-Chart shows the Educational qualification of the Pharmaceutical Representatives, divided into two categories, like Undergraduate and Graduate. The number of Pharmaceutical Representatives is 10, and 40% of the Pharmaceutical Representatives are undergraduate, and the 60% of the Pharmaceutical Representatives are Graduates. However, we can see, the number of Graduates of the Pharmaceutical Representatives is more than Undergraduates of the Pharmaceutical Representatives.



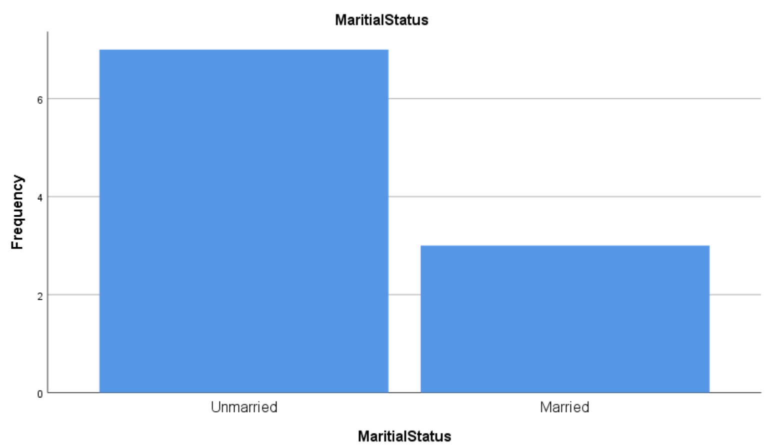
The Pie-Chart shows the Marital Status of the Pharmaceutical Representative, divided into two categories, Like Unmarried and Married. The number of Pharmaceutical Representatives are 10, and 70% of the Pharmaceutical Representatives are Unmarried, and 30% of the Pharmaceutical Representatives are Married. However, we can see, the number of Unmarried of the Pharmaceutical Representatives is more than Married of the Pharmaceutical Representatives.

Bar Chart

A bar diagram, also known as bar chart, is a form of presentation in which the frequencies are presented by rectangles usually separated along the horizontal axis and drawn as bars of convenient widths.



Here the bar chart shows the number of graduate and undergraduate person of pharmaceutical representatives. The bar chart shows the frequency at the vertical axis and education is on the horizontal axis. We can see on the chart, the number of graduate representatives are 4. On the other hand, the number of undergraduate representatives are 8. The graphs simply define the number of undergraduate representatives is bigger than the graduate representatives.



The bar-chart also shows the number of married and unmarried representatives. On this chart the frequencies are on the horizontal axis and marital status is on the vertical axis. The bar chart also shows that the number of married representatives is 7. On the other hand the unmarried representative is only 4. The graphs simply define the number of married representatives is bigger than the unmarried.

Table 02: Numerical Variable

Characteristics	Mean	Median	Mode	Q1	Q3
Age	32.30	32.00	32.00	30.50	33.50
Total Income	38260.00	36050.00	35000.00	35000.00	41375.00
Monthly Phone Ex.	556.00	500.00	500.00	345.00	550.00
Monthly Travelling Ex.	5460.00	5300.00	5000.00	5000.00	6250.00
Monthly Family Ex.	21800.00	20000.00	20000.00	20000.00	25000.00

The following table deals with the Numerical Variable of 10 Pharmaceutical Representative's age, monthly total income, phone expenditure, travelling expenditure, family expenditure and there mean, median, mode, Q1 and Q3. The mean of Age is 32.30, median 32.00 mode 32.00, 1st Quartile 30.50 and 3rd Quartile 33.50.

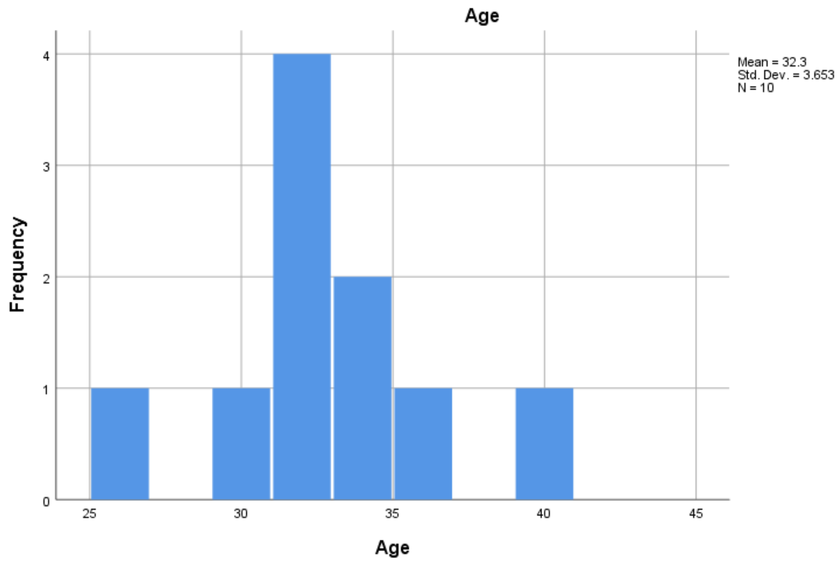
The mean of total monthly income is 38260.00, median 36050.00, Mode 35000.00, 1st Quartile 35000.00 and 3rd Quartile 41375.00. The mean of monthly phone expenditure 556.000000, median 500.00, Mode 500.00, 1st quartile 345.00 and 3rd Quartile 550.00.

The mean of monthly travelling expenditure 5460.00, Median 5300.00, Mode 5000.00, 1st Quartile 5000.00, and 3rd Quartile 6250.00. At last, the mean of monthly family expenditure 21800.00, median 20000.00, Mode 20000.00, 1st quartile 20000.00 and 3rd quartile 25000.00

Histogram

Age

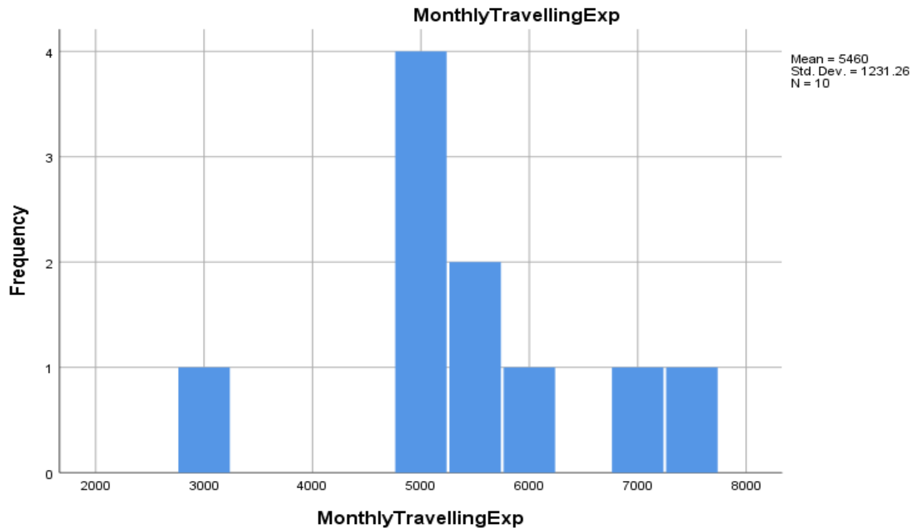
A histogram is constructed by placing the class boundaries on the horizontal axis of a graph and the frequencies on the vertical axis. Each class is shown on the graph by drawing rectangle whose base is the class boundary (Age) and whose height is the corresponding frequency (Number of Pharmaceutical Representative) for the class.



The Histogram shows the highest number of Pharmaceutical Representative is 30-35 years old. The lowest number of Pharmaceutical Representative is 25-30 and 35-40. We can see the 4 Pharmaceutical Representative’s age is 32 and that is the highest number of Pharmaceutical Representative.

Monthly Travelling Exp.

A histogram is constructed by placing the class boundaries on the horizontal axis of a graph and the frequencies on the vertical axis. Each class is shown on the graph by drawing rectangle whose base is the class boundary (Monthly Travelling Ex.) and whose height is the corresponding frequency (Number of Pharmaceutical Representative) for the class.

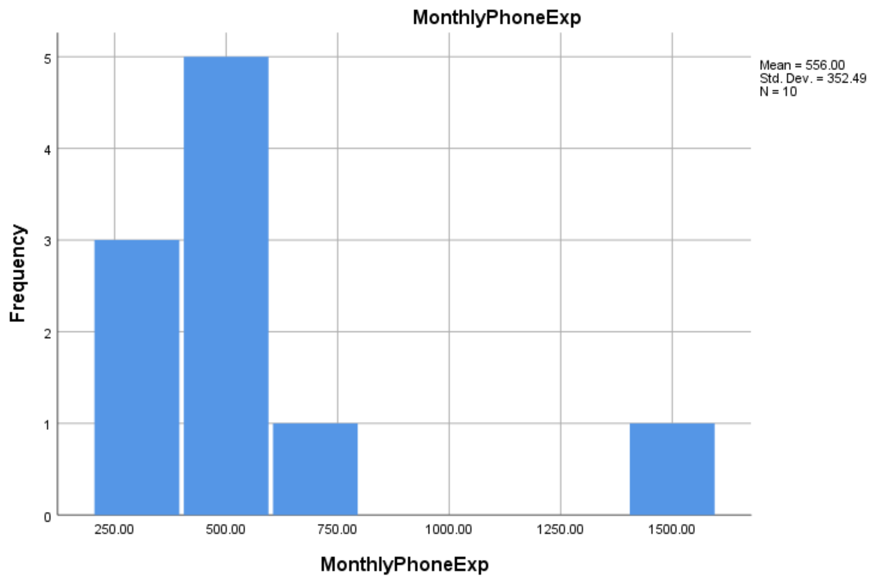


The Histogram shows the highest amount of Pharmaceutical Representative' Monthly Travelling Expenditure is 5000-6000.

The lowest amount of Pharmaceutical Representative's is 2000-3000. We can see the 4 Pharmaceutical Representative's age is 32 and that is the highest number of Pharmaceutical Representative.

Monthly Phone Exp.

A histogram is constructed by placing the class boundaries on the horizontal axis of a graph and the frequencies on the vertical axis. Each class is shown on the graph by drawing rectangle whose base is the class boundary (Monthly Phone Ex.) and whose height is the corresponding frequency (Number of Pharmaceutical Representative) for the class.



The Histogram shows the highest amount of Pharmaceutical Representative' Monthly Phone Expenditure is 250-500.

The lowest amount of Pharmaceutical Representative's is 750-1500. We can see the 5 Pharmaceutical Representative's phone expenditure is 500 and that is the highest number of Pharmaceutical Representatives.

Table 03: Bivariate Table

Characteristic	Monthly Income	Monthly Family Ex.	Monthly Travelling Ex	Monthly Phone Ex.
Education				
• Undergraduate	39875.0±8066.1	22000.0±2449.4	5125.0±1842.7	550.0±100.0
• Graduate	37183.3±2713.2	21666.6±2581.9	5683.3±735.9	560.0±466.4
Marital Status				
• Unmarried	39014.2 ±5440.7	32351.4±3774.8	5585.7 ± 719.8	408.5 ± 92.2
• Married	36500.0±5408.3	32860.0±4131.4	5166.6 ± 2254.6	900.0 ± 529.1

Bivariate Distribution: The distribution in which we consider two variables simultaneously for each item of the series is known as bivariate distribution. The distribution of heights and weights of a group of persons, the ages of husbands and wives of a number of couples, etc. are the examples of bivariate distribution.

The following Bivariate table deals with the **Categorical variable**, like Education (Undergraduate and Graduate) and Marital status (Unmarried and Married) **with Numerical variable**, like Monthly Income, Monthly Family Exp., Monthly Travelling Exp., Monthly Phone exp. In this table we are going to show the comparison of Categorical variable and Numerical variable with **Mean** and **Standard deviation.**

The Undergraduate Pharmaceutical Representative's Monthly income,- mean is 39475 and monthly standard deviation is 8066.1; monthly family expenditure's mean is 22000 and monthly family expenditure standard deviation is 2449; monthly travelling expenditure's mean is 5125 monthly travelling expenditure standard deviation 842.7; after that, monthly phone expenditure's men is 550 and monthly phone expenditure's standard deviation is 100.

The Graduate Pharmaceutical Representative's Monthly income,- mean is 37183.3 and standard deviation is 3713.2; Monthly family expenditure's mean is 21666.6 and standard deviation is 2581.9; Monthly travelling expenditure's mean is 5683.3 and standard deviation 735.9; and Monthly phone expenditure's mean is 560.0 and standard deviation 466.4.

The Unmarried Graduate Pharmaceutical Representative's Monthly income,- mean is 394014.2 and standard deviation is 5440.7; Monthly family expenditure's mean is 32351.4 and standard deviation is 3774.8; Monthly travelling expenditure's mean is 5585.7 and standard deviation is 719.8; and Monthly phone expenditure's mean is 408.5 and standard deviation is 92.2.

The Married Graduate Pharmaceutical Representative's Monthly income,- mean is 3500 and standard deviation is 5408.3; Monthly family expenditure's mean is 32460 and standard deviation is 4131.4; Monthly travelling expenditure's mean is 5166.6 and standard deviation is 2254.6 and Monthly phone expenditure's mean 900 and standard deviation is 529.1

Table 04: Correlations

		Monthly Travelling Exp.	Monthly Phone Exp	Monthly Personal Exp	Total Monthly Exp	Monthly Family Exp	Total Monthly Income	Age
Monthly Travelling Exp.	Pearson Correlation	1	-.286	-.516	.278	.442	.308	-.335
Monthly Phone Exp.	Pearson Correlation	-.286	1	.285	-.020	-.225	-.212	-.360
Monthly Personal Exp.	Pearson Correlation	-.516	.285	1	.569	.212	.200	.054
Monthly Total Exp.	Pearson Correlation	.278	-.020	.569	1	.790**	.688*	-.101
Monthly Family Exp.	Pearson Correlation	.442	-.225	.212	.790**	1	.298	-.412
Monthly Total Income	Pearson Correlation	.308	-.212	.200	.688*	.298	1	.431
Age	Pearson Correlation	-.335	-.360	.054	-.101	-.412	.431	1

The following table shows correlation among the monthly travelling expenditure, monthly phone expenditure, monthly personal expenditure, monthly total expenditure, monthly family expenditure, monthly total income and age. The degree of linear relationship existing between two or more variables observe from same entity is called correlation. We can observe that, most of the variables with one another of the following table are negatively correlated and most of the relations are weak. The correlation of monthly travelling expenditure and monthly phone expenditure is -0.286, the relation is negatively weak. The correlation of monthly travelling expenditure and monthly personal expenditure is -0.516, the relationship is negatively moderate. The correlation of monthly travelling expenditure and total monthly expenditure is 0.278, the relationship is positively weak. The correlation of monthly travelling expenditure and monthly family expenditure is 0.442, the relationship is positively moderate. The correlation of monthly travelling expenditure and monthly total income is 0.308, the relationship is positively weak. The correlation of monthly travelling expenditure and age is -0.335, the relationship is negatively weak.

The correlation of monthly phone expenditure and monthly personal expenditure is 0.285, the relationship is positively weak. The correlation of monthly phone expenditure and monthly total expenditure is -0.02, the relationship is negatively weak. The correlation of monthly phone expenditure and monthly family expenditure is -0.225, the relationship is negatively weak. The correlation of monthly phone expenditure and monthly total income is -0.212, the relationship is negatively weak. The correlation of monthly phone expenditure and age is -0.360, the relationship is negatively weak.

The correlation of monthly personal expenditure and total monthly expenditure is 0.569, the relationship is positively moderate. The correlation of monthly personal expenditure and monthly family expenditure is 0.212, the relationship is positively weak. The correlation of monthly personal expenditure and total monthly income is 0.200, the relationship is positively weak. The correlation of monthly personal expenditure and age is 0.054, the relationship is positively weak.

The correlation of total monthly expenditure and monthly family expenditure is 0.790, the relationship is positively strong. The correlation of total monthly expenditure and monthly total income is 0.688, the relationship is positively strong. The correlation of total monthly expenditure and age is -0.101, the relationship is negatively weak.

The correlation of monthly family expenditure and total monthly income is 0.298, the relationship is positively weak. The correlation of monthly family expenditure and age is -0.412, the relationship is negatively moderate.

The correlation of total monthly income and age is 0.431, the relationship is positively moderate.

Ans: to the Objectives

1. The mean of total monthly income is 38260.00

2. The median of total monthly income is 36050.00

3. The mode of monthly phone expenditure is 500.00

4. The mode of total monthly income is 35000.00

5. The 3rd quartile of monthly travelling expenditure is 6250.00

6. The 1st quartile of monthly travelling expenditure is 5000.00

7. The correlation between total income and total expenditure is 0.688. The relationship is positively strong.

8. The correlation between travelling expenditure and phone expenditure is - 0.286. The relationship is negatively weak.

9. The correlation between age and phone expenditure is - 0.360. The relationship is negatively weak.

10. There are two category of education: undergraduate and graduate. The mean of monthly family expenditure of undergraduate pharmaceutical representative is 22000.0 and the standard deviation is 2449.4. On the other hand, the mean of monthly family expenditure of graduate pharmaceutical representative is 21666.6 and the standard deviation is 2581.9.

Conclusion

While working on the pharmaceutical representative, we were able to identify occupation based on their educational qualification, monthly total income and monthly other expenditure, like family expenditure, travelling expenditure, phone expenditure. Those who have studied up to graduate are mostly associated with this work. The relationship between monthly income and monthly expenditure is positive and strong. So we can say that pharmaceutical representative income is sufficient for their monthly expenses.